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# Towards Concept-Based Language Models for Digital Phenotyping:

A Window into Users' Mental States

Master's Thesis submitted to the  
Faculty of Informatics of the *Università della Svizzera Italiana*  
in partial fulfillment of the requirements for the degree of  
Master of Science in Computational Science

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January 2026

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I certify that except where due acknowledgement has been given, the work presented in this thesis is that of the author alone; the work has not been submitted previously, in whole or in part, to qualify for any other academic award; and the content of the thesis is the result of work which has been carried out since the official commencement date of the approved research program.

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Lugano, 20 January 2026

# Abstract

Depression represents one of the most severe and widespread mental health challenges worldwide. According to estimates by the World Health Organization, approximately 332 million people are affected globally, and even in high income countries only about one third of individuals with depression receive adequate mental health treatment, partly due to limited access to mental health professionals and the difficulty of deploying scalable screening tools. In this context, the growing availability of user generated text on social media platforms offers a unique opportunity for early and large scale mental health screening.

Artificial intelligence is increasingly applied in psychological assessment, yet its limited interpretability remains a significant barrier to clinical integration. While many machine learning models have demonstrated strong predictive performance for mental health detection tasks, they often operate as black boxes, providing little insight into the reasoning behind their decision process. In psychological and clinical settings, this lack of transparency is particularly problematic, as explainability is essential for trust, ethical accountability, and informed decision making. As a result, despite the urgency for accessible and easy to deploy screening systems, AI based solutions are often met with skepticism and remain underutilized in practice.

Concept bottleneck models have introduced a promising paradigm by incorporating human understandable concepts as intermediate representations, but they often do so at the cost of predictive performance. Recent advancements in Concept Embedding Models aim to alleviate this tradeoff by relaxing the strict information bottleneck, enabling both competitive accuracy and more transparent model behavior.

In this work, we investigate the applicability of concept based learning approaches for mental health screening from social media data, with a primary focus on depression. Using user generated text from Reddit, we explore and compare several modeling strategies, including concept bottleneck models and concept embedding models, as well as different retrieval mechanisms for selecting relevant user posts. Depression related symptoms, derived from clinical manuals, are used to guide intermediate representations, allowing for an analysis of how clinically motivated concepts interact with noisy, real world language.

Our results highlight the practical limitations of strict concept bottlenecks in this setting and suggest that concept embedding models provide a more flexible and robust alternative for handling complex and weakly supervised textual signals, while still preserving interpretability hooks that are meaningful for mental health applications.